

MES COLLEGE OF ENGINEERING - KUTTIPPURAM
DEPARTMENT OF COMPUTER APPLICATIONS
20MCA245 – MINI PROJECT

PROFORMA FOR THE APPROVAL OF THE FIFTH SEMESTER MINI PROJECT

(Note: All entries of the proforma for approval should be filled up with appropriate and complete information. Incomplete proforma of approval in any respect will be rejected.)

Mini Project Proposal No :

Batch:

2026-2027

(Filled by the Department)

1. Title of the Project : **SMART WASTE MANAGEMENT SYSTEM FOR HARITHA KARMA SENA**

2. Name of the Guide : **HYDERALI K**

3. Register Number of the Students : **MES25MCA2001**

4. Date of Submission : **14-07-2026**

5. Student Details (in BLOCK LETTERS)

Name

Roll Number

Signature

ABIDAMOL T K

01

For Office Use Only

Approval Status : Approved / Not Approved

Comments/Suggestions :

SMART WASTE MANAGEMENT SYSTEM FOR HARITHA KARMA SENA

1 Introduction

The Smart Waste Management System for Haritha Karma Sena is a web-based application developed using Python Django. The system helps digitize household waste collection by allowing citizens to submit waste pickup requests online. Haritha Karma Sena members can view and update collection requests, while the admin manages users and monitors waste collection activities. This system reduces manual work and improves the efficiency of waste collection.

2 Motivation

The current waste collection process is mostly manual, making it difficult to manage requests and maintain records. This project provides a simple digital solution to improve communication, reduce paperwork, and organize waste collection effectively.

3 Objective/Problem Statement

The objective of this project is to develop a simple and efficient waste collection management system for Haritha Karma Sena. Citizens can request waste pickup online and track the request status. Haritha Karma Sena members can manage pickup requests and update their status. The admin can manage users and monitor waste collection through a centralized system.

4 Functionalities

The system comprises the following core functional modules to ensure structured workflow mapping:

- User Registration and Login: Secure authentication portals for Citizens, Haritha Karma Sena members, and the Administrator.
- Waste Pickup Request: Allows authenticated citizens to seamlessly log in and submit digital requests for household waste clearance.
- View Pickup Status: Enables citizens to track the lifecycle status of their submitted requests in real-time.
- Haritha Karma Sena Member Dashboard: A dedicated terminal for members to check incoming pending collection duties within their boundaries.
- Update Pickup Status: Empowers field operators to safely transition request states after completing physical collections.
- Admin Dashboard: Central control mechanism to facilitate comprehensive user verification and pickup history logs.

5 Developing Environment

The software development parameters configured for constructing the application prototype include:

- Operating System: Windows 11
- Programming Language: Python
- Framework: Django
- Frontend: HTML, CSS, Bootstrap, JavaScript
- Backend: Django
- Database: MySQL

- IDE: Visual Studio Code

6 Conclusion

The Smart Waste Management System provides a simple and efficient platform for managing household waste collection. It reduces manual work, improves communication between citizens and Haritha Karma Sena, and makes the waste collection process more organized and transparent.

References

- [1] Django Documentation – djangoproject.com
- [2] Python Documentation – docs.python.org/3/
- [3] Bootstrap Documentation – getbootstrap.com